Six Sigma

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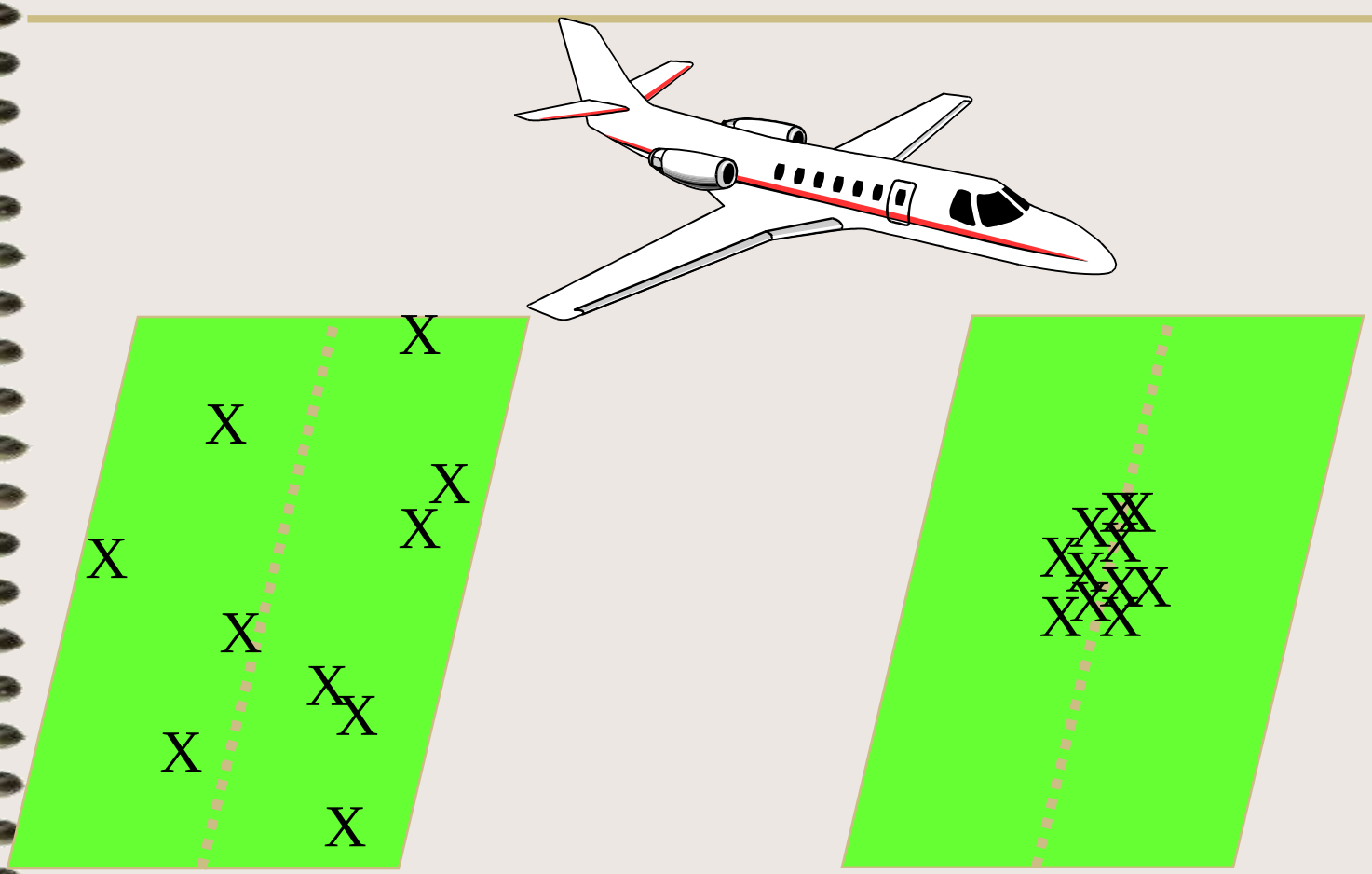
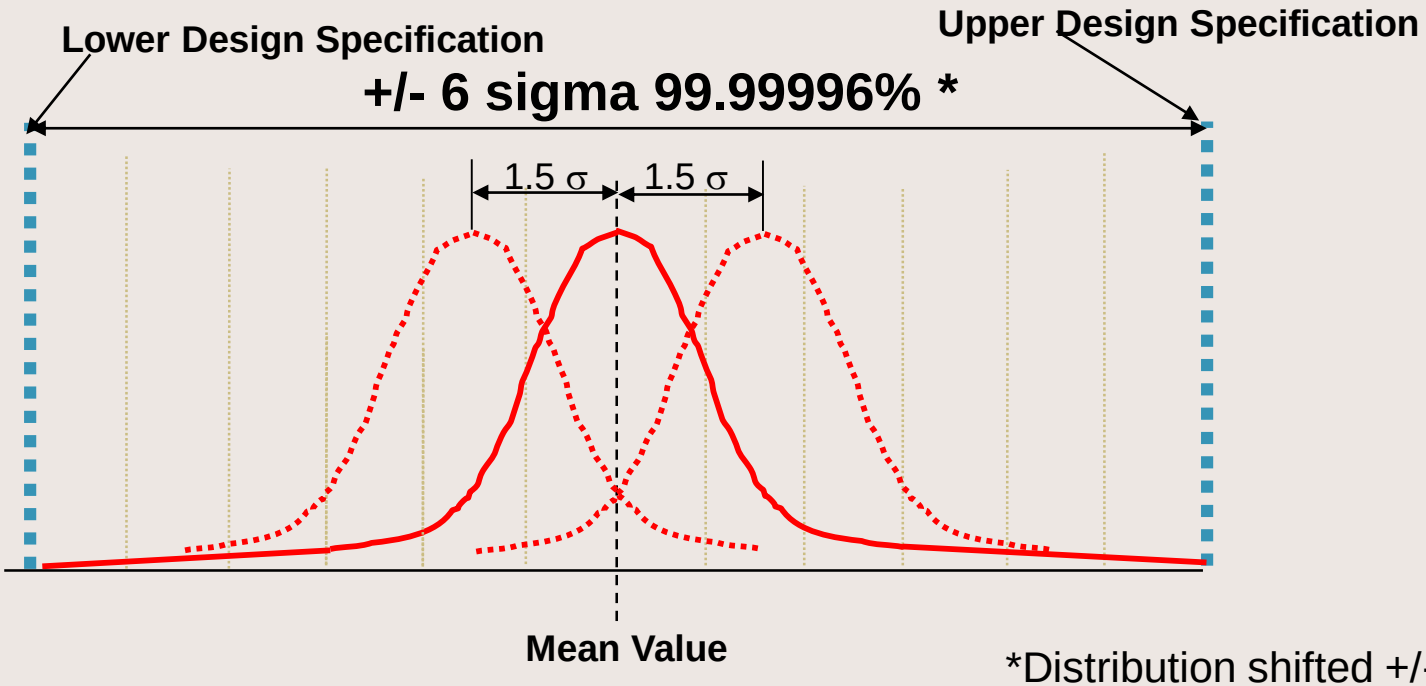
Variation
Which Pilot Would You Choose?



Airplane landing example

Six Sigma Process

- A Six Sigma process is one that has at least six standard deviations between the process mean and the nearest specification limit when centred.



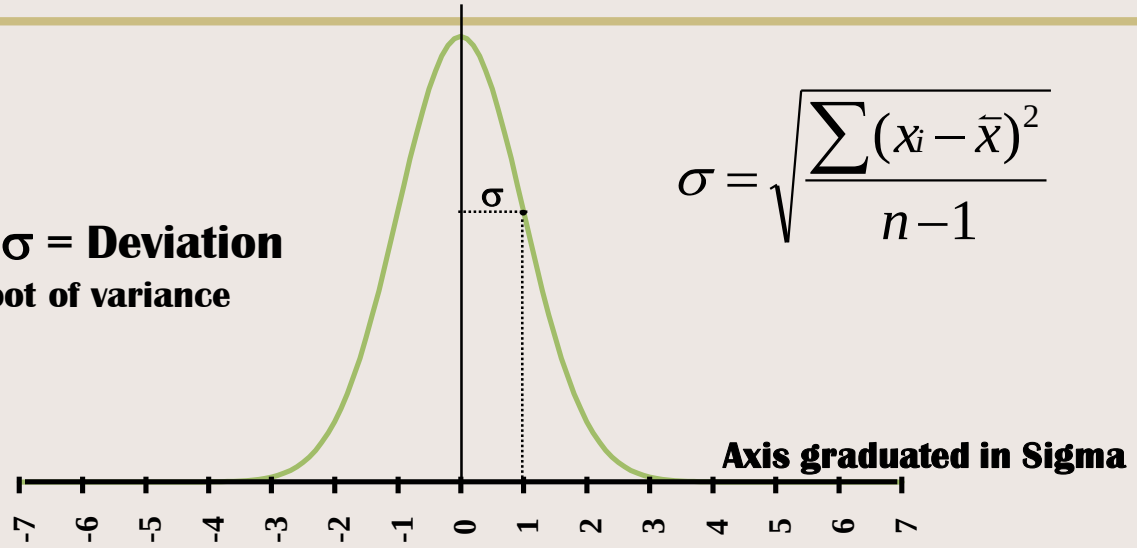
*Distribution shifted +/- 1.5 σ

Six Sigma as a Metric

Sigma = σ = Deviation

(Square root of variance)

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$



between + / - 1σ	68.27 %	317300 ppm outside (deviation)
between + / - 2σ	95.45 %	45500 ppm
between + / - 3σ	99.73 %	2700 ppm
between + / - 4σ	99.9937 %	63 ppm
between + / - 5σ	99.9943 %	57 ppm
between + / - 6σ	99.99966 %	3.4 ppm

Non-Liner Decrease

Industry Average

σ	
1	317,300
2	45,500
3	**2,700**
4	63
5	57
6	3.4

Process Capability

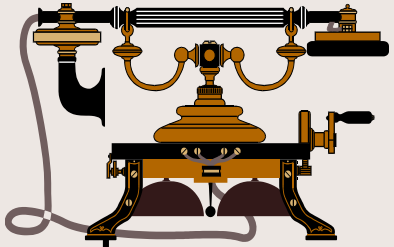
Defects per Million Opportunities

Examples of Sigma

Electric companies if operated at

4 Sigma = no power for 1 hour/week

6 Sigma = no power for 2 seconds/week



Telephone Companies

4 Sigma = No phone service: 4 hours/month

6 Sigma = No phone service: 9 seconds/month

Real-World Differences Between Sigma Levels

99.99% Good (4 Sigma)	99.99966% Good (6 Sigma)
20,000 lost articles of mail/hour	7 lost articles of mail/hour
Unsafe drinking water for 15 minutes every day	1 unsafe minute of drinking water every 7 months
5,000 incorrect surgical operations per week	1.7 incorrect surgical operations per week
200,000 wrong drug prescriptions each year	68 wrong drug prescriptions each year
2 short or long landings at major airports each day	1 long or short landing at major airports every 5 years

History of Six Sigma

- Six Sigma had its birth at Motorola in 1979.
- Six Sigma was developed by Mikel Harry.
- A quantum leap in manufacturing technology occurred at Motorola when it applied Six Sigma.
- Within four years, Six Sigma had saved the company $2.2 billion.

What is Six Sigma ?

- Measure of how well a process is performing
  - Six Sigma process produces 3.4 defects per million opportunities
  - Most companies are at 3-4 sigma level
- Philosophy of reducing defects so we can improve customer satisfaction and reduce costs
- Business Strategy that increases process performance resulting in enhanced customer satisfaction and improved bottom line.

CONCEPT OF SIX SIGMA

- Six-sigma is an approach to continually improve and reduce variations in the process that contribute to waste and poor quality.
- It is a measure of how well a process is performing.
- Six sigma is a well structured, data driven methodology for eliminating defects, waste, or quality control problems of all kinds in every area.
- It Focus on not making mistakes (Mistake proofing) and reducing the variability in the processes.
- Six Sigma properly implemented will
  - Improve customer satisfaction,
  - significantly improve quality,
  - improve the way we launch new products, and
  - create a continuous improvement mindset.

Six Sigma Management Approach

- *"A disciplined method of using extremely rigorous data gathering and statistical analysis to pinpoint sources of errors and ways of eliminating them"*

Harry & Schroeder, 2000

Contd….

Six Sigma Management Approach

- Six Sigma is both a philosophy and a goal
  - Focus on continuous improvement through customer satisfaction
  - Critical processes at 3.4 defects per million opportunities (DPMO)

- The focus is not on DPMO but on a systematic method to reduce variability within the manufacturing process

- Six Sigma projects are lead by individuals selected and trained in the application of this systematic method

- Requires clear definition of roles and responsibilities
  - Green belt, Black belt, Master Black belt, Management Executive

Fundamental Themes of Six Sigma

- Six Sigma is based around a number of central themes
  - A genuine focus on the customer
  - Data and fact driven Management
  - Team based problem solving projects
  - Process focus, management & improvement
  - Proactive management
  - Drive for perfection, tolerance for failure

Contd….

Fundamental Themes of Six Sigma

- A genuine focus on the customer
  - Six Sigma measures start with customer satisfaction
  - Emphasis on understanding customer expectations and requirements
- Data and fact driven Management
  - Speaking with data *(not opinions!)*
  - Management decisions based on fact
  - Develop understanding of internal processes

Contd….

Fundamental Themes of Six Sigma

- Process focus, management & improvement
  - Understanding the process is the key aspect
  - Control the inputs, and improve process

- Proactive management
  - Understanding of Six Sigma principles
  - Defining the root cause of problems
  - Setting and maintaining aggressive improvement targets
  - Challenging ‘why’ things are done this way

Contd….

Fundamental Themes of Six Sigma

- Team based collaboration
  - Six Sigma is teamwork focused
  - Project lead by process expert
  - Facilitated by Six Sigma Black Belt
  - Multi-skill experts within project team
- Drive for perfection, tolerance for failure
  - Being prepared to fail during improvement
  - Understanding why something failed
  - Improving it next time

Six Sigma Supporting Evidence

- General Electric started to use Six Sigma in 1995
  - Total savings estimated to be $15 Billion
  - Over 10,000 GE managers and employees have been trained in Six Sigma methods
- AlliedSignal/Honeywell started Six Sigma improvement in the early 1990s
  - Estimated to be saving over $600 million pa
  - Market value increased by compounded 27% pa
  - *"Its changed the way we think and communicate. We never used to talk about the process or the customer, now they're part of our everyday conversation"*

Six Sigma Supporting Evidence

- Motorola established the first Six Sigma improvement programme in 1987
  - Used to track and compare customer requirements
  - Over the ten years 1987-97
    - Five-fold increase in sales
    - Profits increasing at nearly 20% per year,
    - Share price equal annual increase of over 21% per year
    - Cumulative savings based on Six Sigma at over $14 billion

The Essential Six Sigma Tools & Methods

Statistical Process Control

7 Quality Control Tools

Design of Experiments

Six Sigma

Balanced Scorecards

Process Design/ Redesign

Voice of the Customer

Continuous Improvement

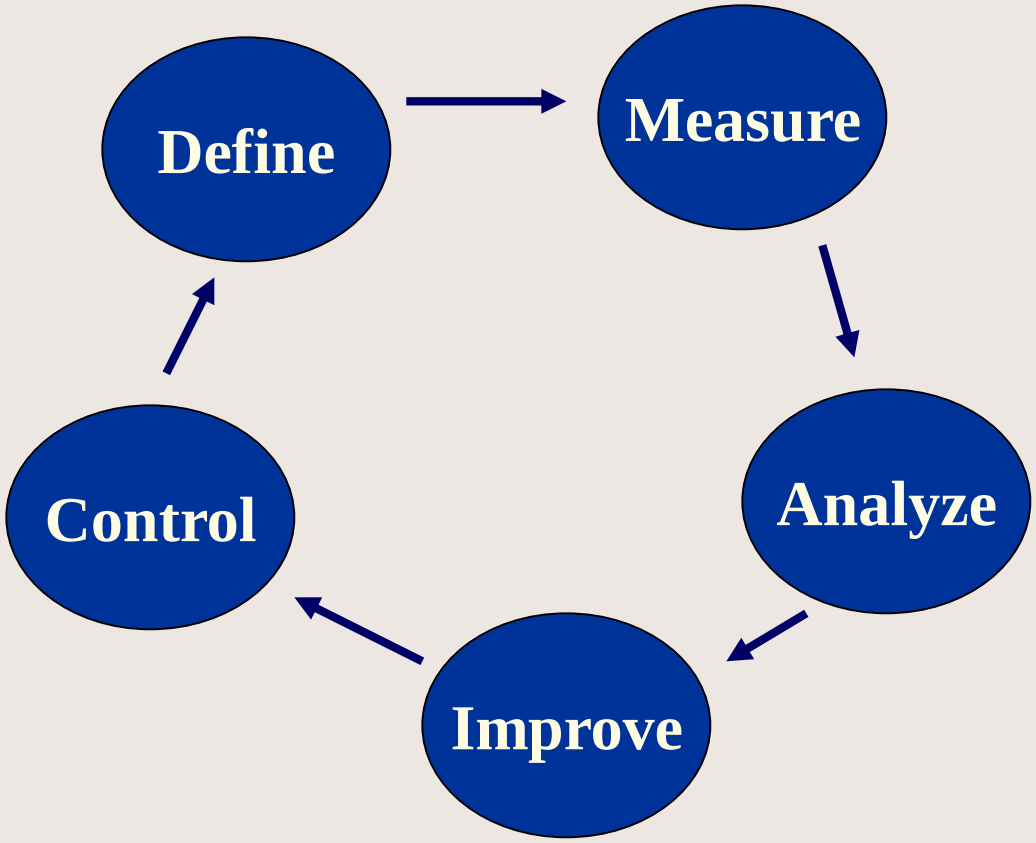
Process Management

Creative Thinking

Six Sigma Improvement Strategy

- Phase 0: Process Definition
- Phase I: Process Measurement
- Phase II: Process Analysis
- Phase III: Process Improvement
- Phase IV: Process Control

DMAIC
Six Sigma Improvement Model



Six Sigma & TQM: Complementary or Conflict?

- **Many of the objectives of TQM are echoed within the Six Sigma approach**
  - **Focus on customers**
  - **Team based activity**
  - **Dependent on people**
  - **Produces an improved business position**

Yet there are significant differences in approach

- **Focused on hard 'process based' deliverables**
- **Precise definition and simple message**
- **Cross functionality of approach**
- **Well defined project time-scales**
- **Well defined project structure**
- **Well defined training objectives**

Benefits of Six Sigma Projects

- Reduced cycle time.
- Increased productivity and added value.
- Improved capacity and output.
- Reduced total defects.
- Increased product reliability.
- Decreased work in progress.
- Improved process flow.
- Faster return on investment.
- Improved overall customer satisfaction

Summary

- Six Sigma
  - It is different
  - It works if you follow the process
  - Produces significant bottom line results
- Change is required by all persons who get involved with the initiative
- Six Sigma will last as long as it produces bottom line results.

Case Study
Bombay Tiffenwallas

Dabbawallas to IIM: Words of bizdom

By Jyothi Prabhakar
TIMES NEWS NETWORK

Their clockwork precision and managerial skills left Prince Charles impressed. Management gurus have observed that the tricks of their trade would do the economy a world of good. Now, these enterprising entrepreneurs have explained to management students, why the bucks stop at the Mumbai *dabbawalla* (lunch-box carrier).

In what marks a first for the B-school script, IIM-Lucknow invited Raghunath Medge and Gangarao Talekar — representatives of the Nutan Mumbai Tiffin Carriers Association — to participate in a seminar attended by students of 50 management institutes across the country.

During the seminar, management students learnt how *dabbawallas* have not only managed to survive in a competitive market, but also expand their business. "I am glad that we have finally been given recognition. I think this has come about becau-se of Prince Charles' response to our work," says Gangaram of the Mumbai Tiffin Box Carriers Association.

Today, the *dabbawalla* has a ticket to everywhere. Because, if it pays to be a *dabbawalla* in Mumbai, the experiment can be replicated nationwide. Whether it is the business of life or the life of business, the money which makes the world go round comes from managing ground reality. That's one thing the *dabbawalla* knows better than the textbook lessons at business school.

delhitimes@indiatimes.com

Case Study
Bombay Tiffenwallas

(Source: Forbes Magazine)

Bombay Tiffanwallas (Raghunath Medge – president)

- They make

one Error on every 16 millions transactions

5000 Tiffanwallas deliver 175,000 lunches everyday and take the empty tiffin back

- They make

one mistake in 2 month

175,000x2(transaction per day) x30 (days)=10,500,000

10,500,000x2 months= 21,000,000 (21 millions)

Case Study
Bombay Tiffenwallas

(Source: Forbes Magazine)

The charge for this service = 150 Rupees per month

Tiffenwallah monthly income= Rs 3,250

Tiffenmen's Association gets Rs 10 per worker

Send to Charity = Rs. 50, 000 per month

Superb!!!

Service and Charity Too

Can anyone ask for more????

SALAM TO THE SPIRIT OF BOMBAY

THANKS!!!